// Position of ID in an array (0 if not found)

1) START
2) FUNCTION FindIndex (Arr : ARRAY[1:5] OF INTEGER, N : INTEGER, ID : INTEGER) RETURNS INTEGER
3) DECLARE i : INTEGER
4) FOR i ← 1 TO N
5) IF Arr[i] = ID THEN
6) RETURN I
7) ENDIF
8) NEXT i
9) RETURN 0
10) ENDFUNCTION

// Input Positive Number of ~~deeds~~ days

11) FUNCTION GetDays () RETURNS INTEGER
12) DECLARE Days : INTEGER
13) REPEAT
14) OUTPUT "Enter number of days"
15) INPUT Days
16) IF Days <= 0 THEN
17) OUTPUT "Days must be positive"
18) ENDIF
19) UNTIL Days > 0
20) RETURN Days
21) ENDFUNCTION

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```
// Making display menu to show customer.
PROCEDURE DisplayMenu()
OUTPUT "1. Register Member"
OUTPUT "2. Display Available Books"
OUTPUT "3. Borrow a Book"
OUTPUT "4. Return a Book"
OUTPUT "5. Display Member Record"
OUTPUT "6. Exit"
OUTPUT "Enter your choice: "
ENDPROCEDURE

// Registering a new member
PROCEDURE RegisterMember (BYREF Name:
STRING, BYREF StudentID : INTEGER,
BYREF MemberType : INTEGER)
OUTPUT "Enter name: "
INPUT Name
OUTPUT "Enter Student ID: "
INPUT StudentID
REPEAT
    OUTPUT "Membership type (1 = Student,
    2 = Faculty): "
    INPUT MemberType
    IF MemberType <> 1 AND MemberType <> 2
    THEN
        OUTPUT "Invalid Membership type"
    ENDIF
```

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42) UNTIL MemberType = 1 OR MemberType = 2

43) ENDPROCEDURE

// Displaying the Books to the member.

44) PROCEDURE DisplayBooks (BookIDs : ARRAY[1:5] OF INTEGER, BookTitles : ARRAY[1:5] OF STRING, AvailableCopies : ARRAY[1:5] OF INTEGER, TotalBooks : INTEGER)

45) DECLARE i : INTEGER

46) OUTPUT "BookID Title AvailableCopies"

47) FOR i ← 1 TO TotalBooks

48) OUTPUT BookIDs[i], BookTitles[i], AvailableCopies[i]

49) NEXT i

50) END PROCEDURE

51) PROCEDURE BorrowBook (BookIDs : ARRAY[1:5] OF INTEGER, BookTitles : ARRAY[1:5] OF STRING, BYREF AvailableCopies : ARRAY[1:5] OF INTEGER, TotalBooks : INTEGER, BYREF BorrowedBookIDs : ARRAY[1:3] OF INTEGER, BYREF BorrowedCount : INTEGER, MemberType : INTEGER)

52) DECLARE ID, i, Days : INTEGER

53) IF BorrowedCount = 3 THEN

54) OUTPUT "Limit Reached: Maximum 3 books at a time"

55) ELSE

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```
56)  OUTPUT "Enter Book ID: "
57)  INPUT ID
58)  i ← FindIndex(BookIDs, TotalBooks, ID)
59)  IF i = 0 THEN
60)      OUTPUT "Invalid Book ID"
61)  ELSE
62)      IF AvailableCopies[i] = 0 THEN
63)          OUTPUT "No Copies of ", BookTitles[i],
                    "available"
64)      ELSE
65)          Days ← GetDays()
66)          BorrowedCount ← BorrowedCount + 1
67)          BorrowedBookIDs[BorrowedCount] ← ID
68)          AvailableCopies[i] ← AvailableCopies[i] - 1
69)          OUTPUT "Borrowed", BookTitles[i]
70)      ENDIF
71)  ENDIF
72)  ENDIF
73)  ENDPROCEDURE
```

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// Function for clients returning a book

74) FUNCTION ReturnBook (BookIDs : ARRAY[1:5] OF INTEGER, BYREF AvailableCopies : ARRAY[1:5] OF INTEGER, TotalBooks : INTEGER, BYREF BorrowedBookIDs : ARRAY[1:3] OF INTEGER, BYREF BorrowedCount : INTEGER, MemberType : INTEGER, BYREF TotalFine : REAL) RETURNS REAL

75) DECLARE ID, Days, Pos, Allowed, Rate, i : INTEGER

76) DECLARE Fine, Payable : REAL

77) DECLARE Code : STRING

78) Payable ← 0

79) IF MemberType = 1 THEN

80) Allowed ← 14

81) Rate ← 20

82) ELSE

83) Allowed ← 30

84) Rate ← 10

85) ENDIF

86) OUTPUT "Enter book ID to return"

87) INPUT ID

88) Pos ← FindIndex (BorrowedBookIDs, BorrowedCount, ID)

89) IF Pos = 0 THEN.

90) OUTPUT "You have not borrowed this book"

91) ELSE

92) Days ← GetDay ()

93) i ← FindIndex (BookIDs, TotalBooks, ID)

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```
94) AvailableCopies[i] ← AvailableCopies[i] + 1
95) IF Days > Allowed THEN
96)     Fine ← (Days - Allowed) * Rate
97)     TotalFine ← TotalFine + Fine
98)     OUTPUT "Late return, Fine charged: Rs.", Fine
99)     OUTPUT "Enter membership code: "
100)    INPUT Code
101)    IF Code = "LIB2026" THEN
102)        Payable ← Fine / 2
103)    ELSE
104)        Payable ← Fine
105)    ENDIF
106)    OUTPUT "Fine payable: Rs. ", Payable
107) ELSE
108)    OUTPUT "Returned on time. No fine"
109) ENDIF
// Remove Return returned book from borrowed list.
110) FOR i ← Pos TO BorrowedCount - 1
111)    BorrowedBookIDs[i] ← BorrowedBookIDs[i+1]
112) NEXT i
113) BorrowedCount ← BorrowedCount - 1
114) ENDIF
115) RETURN Payable
116) ENDFUNCTION
```

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```
117) PROCEDURE DisplayMemberRecord (Name : STRING,
     StudentID : INTEGER, MemberType : INTEGER,
     BorrowedBookIDs : ARRAY[1:3] OF INTEGER, BorrowedCount
     : INTEGER, TotalFine : REAL, FineAfterWaiver : REAL)
118) DECLARE i : INTEGER
119) OUTPUT "Name: ", Name
120) OUTPUT "Student ID: ", StudentID
121) IF MemberType = 1 THEN
122)     OUTPUT "Membership type: Student"
123) ELSE
124)     OUTPUT "Membership type: Faculty"
125) ENDIF
126) IF BorrowedCount = 0 THEN
127)     OUTPUT "Borrowed Books: None"
128) ELSE
129)     FOR i ← 1 TO BorrowedCount
130)         OUTPUT "Borrowed book ID: ", BorrowedBookIDs[i]
131)     NEXT i
132) ENDIF
133) OUTPUT "Total fine before waiver: Rs.", TotalFine
134) OUTPUT "Final fine payable: Rs.", FineAfterWaiver
135) ENDPROCEDURE
```

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```
//Main Program
DECLARE BookIDs, AvailableCopies : ARRAY[1:5] OF INTEGER
DECLARE BookTitles : ARRAY[1:5] OF STRING
DECLARE BorrowedBookIDs : ARRAY[1:3] OF INTEGER
DECLARE Name : STRING
DECLARE StudentID, MemberType, Choice, BorrowedCount : INTEGER
DECLARE Registered : BOOLEAN
DECLARE TotalFine, FineAfterWaiver : REAL
BookIDs ← [101, 102, 103, 104, 105]
BookTitles ← ["Programming in C", "Data Structure", "Discrete Maths", "Database Systems", "Operating System"]
AvailableCopies ← [3, 2, 1, 4, 0]
BorrowedCount ← 0
TotalFine ← 0
FineAfterWaiver ← 0
Registered ← FALSE
REPEAT
    CALL DisplayMenu()
    INPUT Choice
    IF Choice >= 3 AND Choice <= 5 AND Registered = FALSE THEN
        OUTPUT "Please register first"
    ELSE
```

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```
156)        CASE OF Choice
157)            1 : IF Registered = TRUE THEN
                        OUTPUT "A member is already registered"
                    ELSE
                        CALL RegisterMember (Name, StudentID,
                            MemberType)
                        Registered ← TRUE
                    ENDIF
158)            2 : CALL DisplayBooks (BookIDs, BookTitles,
                        AvailableCopies, 5)
159)            3 : CALL BorrowBook (BookIDs, BookTitles,
                        AvailableCopies, 5, BorrowedBookIDs,
                        BorrowedCount, MemberType)
160)            4 : FineAfterWaiver ← FineAfterWaiver +
                        ReturnBook (BookIDs, AvailableCopies, 5,
                        BorrowedBookIDs, BorrowedCount, MemberType,
                        TotalFine)
161)            5 : CALL DisplayMemberRecord (Name, StudentID,
                        MemberType, BorrowedBookIDs, BorrowedCount, TotalFine,
                        FineAfterWaiver)
162)            6 : OUTPUT "Goodbye!"
163)        OTHERWISE : OUTPUT "Invalid choice. Enter 1-6"
164)        ENDCASE
165)    ENDIF
166)    UNTIL Choice = 6
```

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